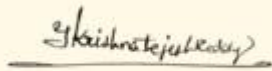


SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:	Y>Krishna Tejesh Reddy	Reg. No.:	192411200
Date:	3-8-2026	Duration:	60 minutes

Code of Conduct

I _Y.Krishna Tejesh Reddy (192411200)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Campus Placement and Recruitment Management Portal

1. Study and Analyze the Scenario

- **Carefully understand the given problem statement.**

The Campus Placement and Recruitment Management Portal belongs to the **Education and Recruitment Management** domain. It provides a centralized platform where students, placement officers, and recruiters can interact efficiently during the campus recruitment process.

- **Identify the business domain, stakeholders, existing challenges, and system objectives.**

Stakeholders

- Students
- Placement Officers

- Recruiters/Companies
- College Administration
- System Administrator

Existing Challenges

- Manual maintenance of student records.
- Difficulty in communicating placement opportunities.
- Delays in scheduling interviews.
- Lack of transparency in application tracking.
- Paper-based verification process.
- Time-consuming report generation.

System Objectives

- Automate the placement process.
- Enable online job postings and applications.
- Simplify interview scheduling.
- Provide real-time placement status.
- Generate placement reports.
- Improve communication among all stakeholders.

2. Identify Functional and Non-Functional Requirements

- **List all functional requirements required by the proposed system.**

Student Module

- Student registration and login.
- Profile management.
- Resume upload.
- View job notifications.
- Apply for jobs.
- Track application status.
- Receive interview notifications.

Recruiter Module

- Company registration.
- Login authentication.
- Post job openings.
- View eligible students.
- Shortlist candidates.
- Schedule interviews.
- Publish selection results.

Placement Officer Module

- Verify student profiles.
- Approve recruiter registration.
- Manage job postings.
- Schedule campus drives.
- Generate placement reports.
- Send notifications.

Admin Module

- Manage users.
- Manage departments.
- Backup database.
- Monitor system activities.
- Generate reports.
- Manage permissions.

- **Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.**

Performance

- Response time should be less than **3 seconds**.
- Support at least **1000 concurrent users**.

Security

- Secure login authentication.
- Password encryption.
- Role-based access control.
- Data privacy protection.

Reliability

- 99.9% system availability.
- Automatic data backup.
- Error recovery mechanism.

Usability

- Simple and user-friendly interface.
- Easy navigation.
- Mobile responsive design.

Scalability

- Support multiple colleges.
- Handle increasing student records.
- Easy integration with future modules.

Maintainability

- Modular software architecture.
- Easy bug fixing.
- Proper documentation.

Availability

- 24×7 online access.
- Cloud database support.
-

3. Identify Actors and Use Cases

- **Identify all primary and secondary actors interacting with the system.**

Primary Actors

- Student
- Recruiter
- Placement Officer
- Administrator

Secondary Actors

- College Database
- Email Notification Service
- SMS Gateway

- **List the major use cases associated with each actor.**

Student

- Register
- Login
- Update Profile
- Upload Resume
- View Jobs
- Apply Job
- Check Application Status
- View Interview Schedule
- Logout

Recruiter

- Register Company
- Login
- Post Jobs
- Search Students
- Shortlist Candidates
- Schedule Interview
- Publish Results
- Logout

Placement Officer

- Verify Students
- Approve Companies
- Manage Job Drives
- Generate Reports
- Send Notifications

Administrator

- Manage Users
- Manage Database
- View Reports
- Backup System
- Configure Settings

4. Prepare the Software Requirement Specification (SRS)

Develop an SRS document containing:

○ Introduction

The Campus Placement and Recruitment Management Portal is a web-based application developed to simplify the placement process in educational institutions. The system connects students, recruiters, and placement officers through a single platform for efficient recruitment management.

- **Problem Description**

Traditional campus placement systems rely on manual record keeping, physical communication, and paper documentation. These methods consume time, increase errors, and make tracking placement activities difficult. Students often miss job opportunities due to delayed notifications, while recruiters face difficulties in accessing eligible candidates.

- **Scope of the System**

The proposed system provides an online platform for managing the entire campus placement process. It allows students to create profiles and apply for jobs, recruiters to post vacancies and conduct recruitment drives, and placement officers to monitor all placement activities efficiently.

- **Functional Requirements**

1. Student Registration
2. Login Authentication
3. Resume Upload
4. Job Posting
5. Job Application
6. Interview Scheduling
7. Candidate Shortlisting
8. Notification Management
9. Report Generation

- **Non-Functional Requirements**

- ❖ High Performance
- ❖ Strong Security
- ❖ Reliability
- ❖ Scalability
- ❖ Availability
- ❖ Maintainability
- ❖ User-Friendly Interface

- **Use Case Descriptions**

Student Login

Actor: Student

Precondition: Student must be registered.

Main Flow

1. Student enters username and password.
2. System validates credentials.
3. Dashboard is displayed.

Postcondition

Student successfully logs into the portal.

- **Data Requirements**

Actor: Student

Precondition

Student profile must be completed.

Main Flow

1. Student views available jobs.
2. Selects a job.
3. Clicks Apply.
4. System stores application.

5. Confirmation message is displayed.

- External Interface Requirements

- User Interface**

- Responsive web interface.
 - Dashboard for all users.
 - Simple navigation menu.

- Hardware Interface**

- Computer
 - Laptop
 - Smartphone

- Software Interface**

- Web Browser
 - MySQL Database
 - Web Server
 - Email Server

- Communication Interface**

- Internet connection
 - Email Notification
 - SMS Notification

- **System Features**

- ❖ Student Registration
 - ❖ Resume Management

- ❖ Job Posting
- ❖ Online Applications
- ❖ Interview Scheduling
- ❖ Placement Dashboard
- ❖ Notifications
- ❖ Report Generation
- ❖ User Management

5. Identify Assumptions and Constraints

- ❖ Students possess valid college IDs.
 - ❖ Recruiters provide genuine company information.
 - ❖ Internet connection is available.
 - ❖ Users have basic computer knowledge.
 - ❖ The database server is continuously available.
 - ❖ Email and SMS services function properly.
- **Identify technical, operational, economic, legal, and time-related constraints affecting the project.**

Technical Constraints

- Requires stable internet connectivity.
- Compatible with modern web browsers.
- Depends on MySQL database.

Operational Constraints

- Users must follow institutional placement rules.
- Placement officer approval is mandatory.

Economic Constraints

- Limited project budget.

- Hosting and maintenance costs.

Legal Constraints

- Student data must remain confidential.
- Compliance with institutional privacy policies.

Time Constraints

- Development must be completed within the academic semester.
- Testing should be finished before deployment.

6. Validate Requirement Completeness and Consistency

- **Verify whether all stakeholder requirements have been addressed.**

Completeness

- All major stakeholders have been identified.
- Functional and non-functional requirements are included.
- Use cases cover all important system operations.

Consistency

- Requirements do not conflict with each other.
- All modules are integrated logically.

Ambiguity Check

- Requirements are written clearly and precisely.
- Technical terms are consistently used.

Missing Requirements Review

- Notification module included.
- Security module included.
- Report generation included.
- Backup and recovery included.

Suggested Improvements

- Integrate AI-based resume screening.
- Add chatbot support.
- Enable online aptitude tests.
- Provide mobile application support.
- Add analytics dashboard for placement statistics.
- Integrate video interview functionality.